

## Program 8

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int i;
    int *a, *b, *c;

    // malloc
    a = (int *)malloc(3 * sizeof(int));

    for (i = 0; i < 3; i++)
        a[i] = i + 1;

    printf("Memory allocated using malloc()\n");

    // calloc
    b = (int *)calloc(3, sizeof(int));

    printf("Memory allocated using calloc()\n");

    // realloc
    a = (int *)realloc(a, 5 * sizeof(int));

    a[3] = 4;
    a[4] = 5;

    printf("Memory increased using realloc()\n");

    // Another allocation
    c = (int *)malloc(2 * sizeof(int));

    printf("Another memory block allocated\n");

    // Free memory
    free(a);
    free(b);
    printf("a and b memory freed\n");

    // Intentionally NOT freeing c
    // This creates a memory leak

    return 0;}
```

## Output

```
[ubuntu@ubuntu:~/OSLab/Practical]$ nano program8.c
[ubuntu@ubuntu:~/OSLab/Practical]$ gcc program8.c -o program8
[ubuntu@ubuntu:~/OSLab/Practical]$ ./program8
Memory allocated using malloc()
Memory allocated using calloc()
Memory increased using realloc()
Another memory block allocated
a and b memory freed
ubuntu@ubuntu:~/OSLab/Practical$
```